

# Literature Reading List

IITB–Optiver AI Innovation Lab — Projects 1 & 2 All arXiv links have a free "Download PDF" button on the page.

Read these first (directly cited on your slides)

| Paper                                                                                                    | Why it matters                                                                      | Link                                                                            |
|----------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| Gorishniy et al., <i>Revisiting Deep Learning Models for Tabular Data</i> (FT-Transformer), NeurIPS 2021 | The primary architecture on slide 4                                                 | <a href="https://arxiv.org/abs/2106.11959">https://arxiv.org/abs/2106.11959</a> |
| Somepalli et al., <i>SAINT: Improved Neural Networks for Tabular Data via Row Attention</i> (2021)       | The reserve architecture for fraud tasks                                            | <a href="https://arxiv.org/abs/2106.01342">https://arxiv.org/abs/2106.01342</a> |
| Huang et al., <i>TabTransformer</i> (2020)                                                               | The architecture you deprioritised — know why                                       | <a href="https://arxiv.org/abs/2012.06678">https://arxiv.org/abs/2012.06678</a> |
| Arik & Pfister, <i>TabNet: Attentive Interpretable Tabular Learning</i> (2019)                           | One of your three baselines                                                         | <a href="https://arxiv.org/abs/1908.07442">https://arxiv.org/abs/1908.07442</a> |
| Snell et al., <i>Prototypical Networks for Few-shot Learning</i> (2017)                                  | Core of theme 1.2                                                                   | <a href="https://arxiv.org/abs/1703.05175">https://arxiv.org/abs/1703.05175</a> |
| Lin et al., <i>Focal Loss for Dense Object Detection</i> (2017)                                          | The imbalance-aware loss for theme 1.5                                              | <a href="https://arxiv.org/abs/1708.02002">https://arxiv.org/abs/1708.02002</a> |
| Padhi et al., <i>Tabular Transformers for Modeling Multivariate Time Series</i> (TabBERT, IBM, 2020)     | BERT-style masked pretraining on transaction logs — closest prior work to Project 2 | <a href="https://arxiv.org/abs/2011.01843">https://arxiv.org/abs/2011.01843</a> |

| Paper                                                           | Why it matters           | Link                                                                          |
|-----------------------------------------------------------------|--------------------------|-------------------------------------------------------------------------------|
| Kingma & Welling, <i>Auto-Encoding Variational Bayes</i> (2013) | The VAE behind theme 1.4 | <a href="https://arxiv.org/abs/1312.6114">https://arxiv.org/abs/1312.6114</a> |

## Project 1 — sparse, irregular, imbalanced data

### 1.1 Sparse / irregular event sequences

- Shukla & Marlin, *Multi-Time Attention Networks for Irregularly Sampled Time Series* (mTAN, ICLR 2021) — <https://arxiv.org/abs/2101.10318>
- Zhang et al., *Raindrop: Graph-Guided Networks for Irregularly Sampled Time Series* (ICLR 2022) — <https://arxiv.org/abs/2110.05357>
- Che et al., *Recurrent Neural Networks for Multivariate Time Series with Missing Values* (GRU-D, 2016) — <https://arxiv.org/abs/1606.01865>
- Chen et al., *ContiFormer: Continuous-Time Transformer for Irregular Time Series Modeling* (NeurIPS 2023) — <https://arxiv.org/abs/2402.10635>

### 1.3 Missing modalities

- Wu et al., *Deep Multimodal Learning with Missing Modality: A Survey* (2024, updated Feb 2026) — <https://arxiv.org/abs/2409.07825>
- Ma et al., *Are Multimodal Transformers Robust to Missing Modality?* (CVPR 2022) — <https://arxiv.org/abs/2204.05454>

### 1.5 Class imbalance & fraud

- *A Survey of Deep Long-Tail Classification Advancements* (2024) — <https://arxiv.org/abs/2404.15593>
- *Evaluating Supervised Learning Models for Fraud Detection on Imbalanced Transaction Data* (2025) — <https://arxiv.org/abs/2505.22521>
- *Financial Fraud Detection via Deep Learning: A Systematic Literature Review* (2025) — <https://arxiv.org/abs/2502.00201>

## Project 2 — foundation models for tabular & time-series data

### Tabular foundation models

- Hollmann et al., *TabPFN* (ICLR 2023) — <https://arxiv.org/abs/2207.01848> (v2 is published in Nature, Jan 2025: "Accurate predictions on small data with a tabular foundation model")

- *A Closer Look at TabPFN v2: Strength, Limitation, and Extension* (2025) — <https://arxiv.org/abs/2502.17361>
- *TabICLv2: A Better, Faster, Scalable, and Open Tabular Foundation Model* (2026) — <https://arxiv.org/abs/2602.11139>

### Transaction / financial foundation models (closest to your project)

- Skalski et al., *Towards a Foundation Purchasing Model: Pretrained Generative Autoregression on Transaction Sequences* (2024) — <https://arxiv.org/abs/2401.01641>
- *Open Banking Foundational Model: Learning Language Representations from Few Financial Transactions* (2025) — <https://arxiv.org/abs/2511.12154>
- *Enhancing Foundation Models in Transaction Understanding with LLM-based Sentence Embeddings* (2026) — <https://arxiv.org/abs/2601.05271>

### Time-series foundation models

- Ansari et al., *Chronos: Learning the Language of Time Series* (Amazon, 2024) — <https://arxiv.org/abs/2403.07815>
- Das et al., *A Decoder-Only Foundation Model for Time-Series Forecasting* (TimesFM, Google, 2023) — <https://arxiv.org/abs/2310.10688>
- Woo et al., *Unified Training of Universal Time Series Forecasting Transformers* (Moirai, Salesforce, 2024) — <https://arxiv.org/abs/2402.02592>
- Goswami et al., *MOMENT: A Family of Open Time-Series Foundation Models* (2024) — <https://arxiv.org/abs/2402.03885>
- *Foundation Models for Time Series Forecasting* (survey, 2025) — <https://arxiv.org/abs/2507.08858>

### Suggested reading order

1. FT-Transformer, SAINT, TabTransformer abstracts + intro (30 min) — you must be able to defend the architecture choice on slide 4.
2. TabBERT + Foundation Purchasing Model (1 hr) — the two closest prior works; expect Optiver to know these.
3. TabPFN v2 commentary paper (30 min) — likely to come up if anyone says "hasn't this been done?"; your answer: TabPFN targets small datasets, not large-scale financial pretraining.
4. Skim the two surveys (missing modality, long-tail) for the related-work vocabulary.